



Advancing Sustainable Energy Transition in Brazil Through Renewable Energy Expansion and Waste Valorisation: Challenges, Opportunities and Policy Pathways

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Abstract: The transition towards low-carbon energy systems has been increasingly recognised as a critical global priority for mitigating climate change, reducing dependence on fossil resources, and promoting sustainable socioeconomic development. Although Brazil possesses one of the world's most renewable energy matrices, supported primarily by hydropower, bioenergy, wind, and solar resources, significant challenges remain in the effective integration of waste-to-energy technologies and circular resource management strategies. In this review, the current status of renewable energy deployment in Brazil is critically assessed alongside the generation, management, and valorisation potential of major waste streams, including agricultural biomass, agro-industrial residues, food-processing wastes, municipal solid waste (MSW), wastewater sludge, construction and demolition waste (CDW), pulp and paper residues, and end-of-life tyres. Existing treatment practices and recovery technologies are systematically examined with emphasis on their capacity to convert waste into value-added products such as solid, liquid, and gaseous biofuels, secondary raw materials, and platform chemicals. Particular attention is given to technological limitations, regional disparities in infrastructure, and policy gaps that have constrained the broader implementation of decentralised renewable energy systems, especially in rural and residential sectors. It is observed that despite substantial progress in renewable electricity generation, waste recycling rates, energy recovery efficiency, and integrated waste management practices remain comparatively underdeveloped. The adoption of advanced thermochemical, biochemical, and material recovery technologies is shown to offer significant opportunities for emissions reduction, resource efficiency, and industrial symbiosis. Furthermore, the role of regulatory frameworks, economic incentives, and public investment in accelerating the transition towards a circular and low-carbon economy is highlighted. The findings demonstrate that the strategic integration of renewable energy expansion with waste valorisation could substantially enhance energy security, environmental performance, and economic resilience in Brazil. The technological pathways, policy mechanisms, and management strategies discussed are also considered transferable to other emerging economies facing similar energy and waste management challenges.

Keywords: Energy transition; Biofuels; Agriculture and food wastes; Pulp and paper wastes; Construction and demolition waste; Discarded tires

1 Introduction

The intense use of petroleum-based energy, continuous aggression on the environment, and harming the ecosystem by generating and depositing waste in inadequate areas, besides deforesting and malusing land and water, among other actions, led to frequent and severe catastrophes in many regions around the world. Energy transition is the logical way to limit the global warming effects and reduce the environmental impacts and loss of ecosystems and threats to food and energy securities. Sustainable bioenergy, preservation of forests, better management of cultivated lands, and globally incorporating more renewable energy in the power systems are among the actions in line with the Paris Accord 2015 [1].

Decarbonization, decentralization and democratization are among the main pillars for energy transition involving energy generation, consumption and efficient use. The transition not only includes replacing fossil-based energy with

clean renewable energy from wind, solar and biomass sources, but also extends to valorization of residues, adequate waste treatment, minimization of all types of losses, energy efficiency, and better use of natural resources.

A general analysis shows that the transport sector, specifically shipping and aviation, made modest efforts to adopt low-carbon technologies, while bioethanol and biodiesel are fueling road transport. The combination of sustainable biofuels and advanced technologies can reduce the annual CO₂ emissions. There is widespread agreement to reduce CO₂ emissions from transport by 50% until 2050 [2].

The building sector is recognized as a big indirect consumer of electricity and a strong indirect contributor to global emissions due to its big consumption of steel and cement. Global efforts in research and development must be done to decarbonize the sector and reduce its energy consumption and emissions.

The industrial sector is not only a big consumer of petroleum-based fuels, but also a big consumer of electricity. Industry currently contributes about 9 Gt of CO₂ annually. To reach 2050 net-zero goals, industrial emissions must drop by over 70% [3]. This can be achieved by adequate management of energy use, use of renewable energies, minimizing thermal and materials losses besides investing in new technologies that promote efficiency and productivity.

Although Brazil has a diversified energy matrix, more inclusion of distributed renewables must be encouraged by adequate public policies and financial incentives. Many solar and wind energy farms are already incorporated in the Brazilian energy matrix fulfilling the necessity of concentrated power generation. In relation to distributed generation, efforts are still modest but must be increased by adequate public policies to meet the basic pillars for sustainable energy transition which require decentralization and democratization of energy generation and use. The sustainable energy transition must not be limited to implementing renewables in the energy sector but must go far ahead and intensify distributed energy generation and use for the residential sector, rural areas, and isolated communities. Actions must include improvement of sanitation, increase of recycling, strengthen circular economy, besides eliminating losses of food, water, raw material and energy.

The objective of this study is to map the renewables in Brazil, including solar and wind energy, and the actual contribution of organic wastes such as biomass, agriculture and food processing to biofuel production. Other relevant wastes from the construction and demolition sector, pulp and paper industry, and discarded tires are also addressed to recover materials, energy, and enhance circular economy. External experience and available technologies are presented to help enhance future changes and new routes for improved utilization of current and future resources.

The main contribution of the present review is that it shows the potential and challenges facing energy transition in Brazil, indicating the gaps and possible opportunities for research and developments to achieve a sustainable energy transition that embraces the three sustainability pillars. The Brazilian current situation was mapped, followed by possible contributions from the renewables sector with a focus on distributed generation and consumption. Contributions of the residues from the agriculture and food processing industries and other highly polluting sectors such as the paper and pulp industry, tires and demolition and building industry were also accessed to direct necessary efforts mainly to strengthen the social and ambient pillars without forgetting the financial pillar. To our knowledge, this review with this specific approach is new and never published before.

2 About Brazil

Considering the actual scenario of electricity generation in Brazil, new projects for hydraulic electricity generation are not welcome due to their destructive impacts on native lands, forests, and natural habitats of populations and flora and fauna. Vulnerability of the hydraulic was demonstrated due to lack of sufficient rains during the last years to recover water used in the hydroelectric energy generation. This seriously threatened the energy security of the country and urged diversification of the energy resources and more inclusion of wind and solar photovoltaic energy farms. Due to the big incentives and the favorable acceptance of the population, photovoltaic electricity generation presented a strong evolution with an increasing rate of residential and small commercial installations. The increase of these small installations represents a positive contribution to democratic access to renewable energy generation and use, which is essential for energy transition and in line with the UN recommendations.

The Brazilian wind potential can achieve about 1.3 TW, which strengthens energy security and reduces emissions since each MW installed can reduce 23 Mt per year in CO₂ emissions. It is important to remember that new challenges will appear when wind rotors reach the end of their working life and need to be discarded without causing big impacts on the environment [4].

Brazil is a big producer of agricultural crops, biomass and biofuels, mainly ethanol and biodiesel. Sugarcane, agricultural waste, and sugarcane bagasse are among the main sources for producing ethanol, while biodiesel production is sustained mainly on a food crop, soybean, used oils and animal fats, which currently are not enough to sustain the internal consumption, which is about 75% of the national internal demands. Without needing to change land use or invade the food resources, it is essential to investigate available agriculture and food waste and implement new technologies to extract materials from these wastes, besides improving the existing technology for converting biomass to energy and using municipal solid waste (MSW) and other wastes for generating electric energy and producing liquid and gaseous fuels.

Agricultural and food wastes are crucial for the energy transition, providing sustainable feedstocks for biofuels (like bioethanol, biodiesel, biogas/biomethane) via biological, chemical, and thermochemical processes (fermentation, anaerobic digestion, pyrolysis, gasification), reducing fossil fuel dependence, greenhouse gas (GHG) emissions, and enhancing the circular economy. In 2019, Brazil produced 5.9 million cubic meters of biodiesel and 6.25 cubic meters in 2022, a significant increase but still not sufficient for the internal demands. Currently, Brazil produces bioethanol, both first and second generation, from sugarcane and also from corn. In 2022 the production was about 30.75 billion liters [5].

The available technologies for converting agriculture and food wastes into biofuels include fermentation, anaerobic digestion, pyrolysis, and gasification, which are distinct biomass-to-energy pathways with different optimal feedstocks, efficiencies, and product types. Anaerobic digestion is highly effective for wet organic waste to produce biogas (40–60% methane), while gasification and pyrolysis are thermochemical processes that efficiently convert dry biomass into syngas, bio-oil, and biochar. Gasification and pyrolysis are generally higher-efficiency, faster, and more versatile for dry, complex materials, while anaerobic digestion and fermentation are more efficient for wet, degradable organic waste. Gasification has a high efficiency for conversion of diverse organic waste into usable syngas, with energy recovery efficiencies often around 80%, while pyrolysis is highly efficient for producing bio-oil and char, achieving energy recovery efficiencies up to 90% depending on the process conditions. Anaerobic digestion is efficient for high-moisture organic waste (e.g., sewage sludge, food waste), typically with a net energy recovery around 70%, though this can be increased with integrated technologies. Fermentation generally refers to smaller-scale microbial conversion for specific fuels (like ethanol) or chemicals; however, it is limited by substrate specificity compared to thermochemical methods. Transesterification converts triglycerides (oils/fats) into fatty acid methyl esters (biodiesel) and glycerol by reacting with alcohol (methanol/ethanol) using a catalyst (NaOH/KOH), achieving high efficiency (>95% yield) under optimal conditions. Microwave/ultrasonic methods cut reaction times from hours to minutes [6–9].

Sustainable energy transitions also include the treatment of industrial, agriculture and urban wastes, the elimination of dumping sites, besides intensifying recycling and the circular economy to reduce consumption of raw materials, water and energy and reduces impacts on the ambient and pollution of underground water and soil.

Brazil has a deficit in water and sewage services. According to study [10], in 2020, 84% of Brazilian households had a water supply, 55% had sewage collection, and 51% of the generated sewage received treatment. On average, only 84.9% of the population is currently supplied with drinking water. Current challenges include wide access to treated water in urban and rural areas, and improvement of sewage treatment issues.

Biogas is one of the least used biofuels in Brazil, yet the country has 810 plants, and produced about 2.8 billion cubic meters in 2022. These biogas plants mainly produce electrical energy (86%), thermal energy (10%), mechanical energy (1%) and/or biomethane (3%) [11]. The production of biogas from organic waste is carried out through a process of separation, homogenization and anaerobic biodigestion. In turn, part of the biogas comes from effluents from wastewater treatment plants, which usually use conventional aeration-activated sludge technology, and also from MSW, through a dry methanization process, which can generate biogas and stabilized waste [11, 12]. In support of the energy transition and better and efficient use of available energy resources, biogas is receiving attention from the federal energy authorities through new incentives and regulations relative to the generation and use of biogas. The available literature shows that one ton of organic waste can produce approximately 100 to 300 cubic meters of biogas through anaerobic digestion, or roughly 50 to 150 cubic meters of methane, while mixed MSW averages closer to 50 cubic meters of methane per ton. Brazil generates over 80 million tons of mixed MSW annually. According to 2024 data, this amounted to approximately 81.6 million tons. Brazil has massive potential for energy recovery from mixed MSW, with estimates suggesting that utilizing 47% of urban waste could install roughly 3.3 GW of capacity, enough to power over 4 million homes [11, 13–15].

Globally, over 1.5 billion waste tires are generated annually worldwide. Only about 42% are utilized in material recovery, and 15% for energy recovery, while the rest ends up in landfills. Scrap tires generated in Brazil are about 20 million units, of which some are discarded irregularly in landfills.

The Brazilian mandatory reverse logistics system for disposed tires shows that approximately 90 million units are discarded annually, and the vast majority are collected and sent for energy recovery and material recycling. An average of 40–53 million new tires is produced annually in Brazil, and other millions are imported. Brazil's National Solid Waste Policy (PNRS) and National Council for the Environment (CONAMA) Resolution No. 416/2009 mandate that manufacturers and importers be responsible for implementing reverse logistics systems to collect and provide an environmentally friendly destination for one unserviceable tire for every new one sold on the market [16].

The main destinations for collected waste tires are co-processing for energy recovery and material recycling. Co-processing is the most common destination, accounting for roughly 67–70% of disposed tires, where they are used as an alternative fuel in high-temperature clinker kilns at cement factories. The remaining tires (around 30–33%) are recycled into various products by processes such as granulation and pyrolysis, among others. In the granulation process, tires are cut and ground into rubber powder or granules used in rubber-asphalt for pavements, sports court flooring, synthetic grass, and industrial floors. In the pyrolysis process, rubber is decomposed in the absence of

oxygen to produce oils, steel, and carbon black. In Brazil, despite the well-defined regulatory framework, challenges remain, and more intensive control is needed to avoid tires ending up in landfills or open dumps [16].

The pulp and paper industry generates a huge amount of recyclable waste such as lime mud, ash, and pulp mill sludge, which can be processed and used in agriculture, construction, and energy technology. New policies are required to encourage the recycling of these materials [17–19].

The building sector is a big energy consumer and carbon emitter due to the basic material used, mainly cement and steel. Brazil generates approximately 45–70 million tons of construction and demolition waste (CDW) annually, most of which (bricks, concrete, mortar, etc.) is highly recyclable. This preserves natural resources like sand and gravel, reducing the significant environmental impact and energy consumption associated with extracting and processing virgin materials [20]. Recycling CDW into aggregates for use in public works (e.g., road paving) is a key strategy.

Remarks

The current scenario shows the need for incentives to enhance individual and small photovoltaic (PV) projects to promote decentralization and democratic use of renewable energy by low-income populations. Similar actions are required for wind energy with the increase of urban wind energy projects for condominiums and roof-mounted installations besides investing in offshore farms and initiating plans for receiving wind rotors at their end-of-life operation. Concentrated and organized federal, state, and municipal efforts are necessary to resolve solid waste treatment and eradicate dumps and accelerate recycling and treatment of sewage and wastewaters. There is also a need to invest more in research and technology and improve methods for generating biofuels and chemical products from agricultural and food wastes. More control, regulation, and incentives are required to enhance recycling of pulp and paper waste, CDW, and discarded tires, besides the implementation of awareness programs for schools and communities to alert them against all types of waste, principally food, water and energy.

3 Contribution of Renewables to Energy Transition

The continuous increase of the population, industrialization and improvements in the living standards are among the pressing factors for the increase of global demands for energy. Migration to renewable energy sources seems to be an urgent and viable solution that can help face these challenges [21]. Energy poverty, unreliable power networks and systems, reforms, privatization, and financial instability are crucial challenges to manage energy systems in developing countries [22]. Increasing the share of renewable energy resources and reducing electricity based on fossil fuels can help the decarbonization of the energy sectors [23].

Hassan et al. [24] presented a study that investigates the trajectory of renewable energy implementation, the importance of investments, and collaborations to accelerate the global shift to energy transition. Figure 1 shows the rapid expansion and projection of PV power installed in Brazil until 2025.

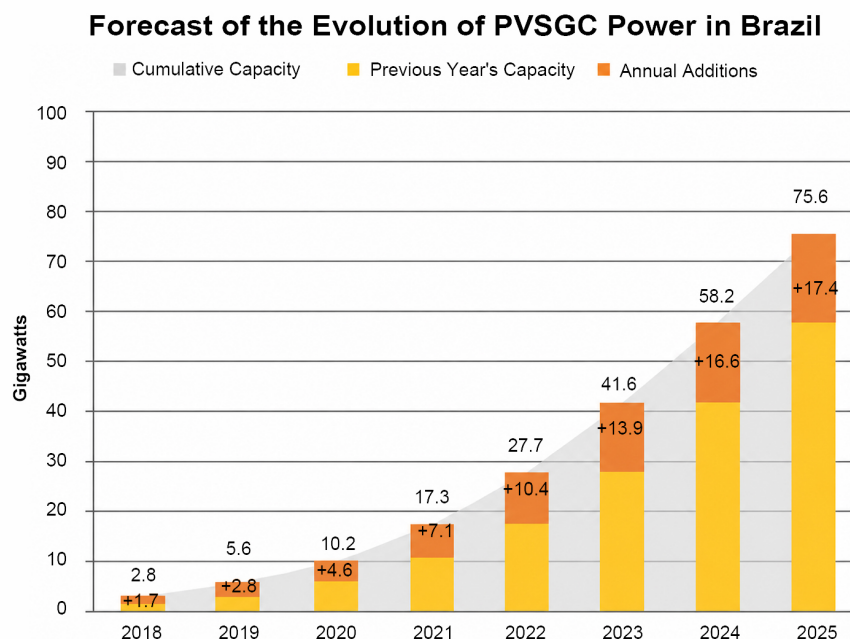


Figure 1. Evolution of the connected photovoltaic (PV) systems until 2025 [25]

Solar energy contributes a small fraction to the global electricity generation, with an eminent need for more investments in PV farms and residential and community installations to strengthen distributed generation and utilization and enhance global generation and energy distribution efficiency [26]. Breyer et al. [27] used scenarios and an energy transition model to investigate the contribution of PV in the global energy transition. The authors commented that the generation capacity could achieve 42 TWp, and that solar PV electricity is expected to be cheaper and widely used for electricity supply.

Parallel to the wide expansion of PV electricity installations, wind energy experienced global expansion during the last three decades to alleviate global warming effects and reduces the dependence on fossil-based fuels. Wind farms for centralized electricity generation were planned worldwide, including developing countries such as Brazil. Şahin [28] reviewed wind energy history, technology, economy, and installed capacity. The status of wind energy development worldwide was presented, and the factors that promote the development of wind projects were discussed.

Most of the Brazilian wind energy farms are localized in regions covered by native vegetation and coastal sands, which cause serious problems of mechanical erosion of blades and internal rotating parts of the generation systems [29, 30]. Future expansion of wind power farms must take this into consideration to avoid sand movements and land degradation. Offshore wind farms must be considered for big machines, while small and medium-capacity machines must be considered for decentralization of generation and use in urban and rural areas [31, 32]. Figure 2 shows the high expansion rate and the tremendous increase of the installed wind energy capacity in the period 2005–2024 [33].

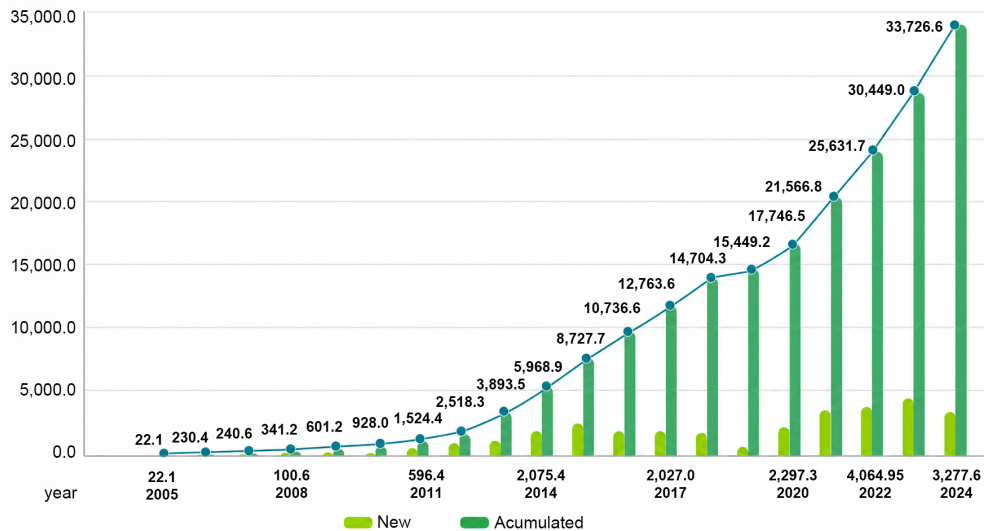


Figure 2. Installed capacity of wind power in Brazil [33]

Brazil depends heavily on hydraulic electric energy generation for supplying the national distribution grid which makes the national distribution system sensitive to environmental conditions, duration and intensity of rainy seasons and accumulated water reserves behind the different dams in the water circuit. The water crisis of 2021 dramatically affected the hydraulic electricity generation and showed the fragility of the hydraulic generation system and exposed the country's vulnerability to energy insecurity. This crisis showed the need for more diversification of the energy supply systems and the participation of diverse renewable sources. Since then, continuously increasing investments are destined to implant solar and wind energy farms, increase the participation of biomass in electricity generation and biofuel production. Recently Petrobras announced interest in financing bioenergy projects to help the transport and industrial sectors achieve independence from the fossil-based fuels and products.

There is a strong need to decentralize solar and wind installations to attend to local demands, isolated and rural areas, individual residences, buildings and residential condominiums in order to improve electricity distribution efficiency, reduce energy losses, and ensure both equity and a secure energy supply [34]. To achieve these objectives, intense investments are needed in the industrial sector to develop small and medium low-noise wind turbines constructed from recyclable materials for use in urban and rural areas and on the roofs of buildings and commercial installations. With reference to solar PV installations, solar PV panels, batteries and other equipment could be designed and produced nationally to attend to the Brazilian ambient conditions besides creating local valuable know-how and jobs.

Remarks

The installed wind energy capacity in Brazil in 2024 was 29 GW, generated by 1027 wind farms mainly connected to the national grid. The number of installed small wind turbines is marginal and restricted to private users and small agricultural farms. The installed solar PV capacity increased from 37.647 GW in 2023 to 50.675 GW in 2024. The small-scale solar PV energy market showed a significant increase in 2024 due to recent regulations and commercial incentives, achieving 8.7 GW mainly from micro- and mini-generation. Besides wind and solar energy, other renewables such as agriculture and food waste can significantly contribute to energy transition by producing biofuels and electricity and reducing emissions and waste sent to landfills.

4 Contribution of Agriculture and Food Wastes

Agricultural wastes are composed of organic and inorganic wastes. The organic part includes crops and animal waste and food byproducts. Inorganic waste includes non-biodegradable materials such as plastics, packaging, and certain agrochemicals. Organic waste is generally composed of plant materials left in fields after harvesting, like stalks, leaves, and straw from crops such as wheat, rice, corn, and sugarcane; livestock waste which includes animal waste, bedding materials, and carcasses; industrial food processing waste such as discarded fruits and vegetables, or processing residues like sugarcane bagasse, rice husk, orange peel, etc. Further, agricultural waste can be categorized based on its source within the agricultural process, including primary (field residues after harvest), secondary (processing and manufacture waste), and tertiary (post-consumer) waste. Food waste represents a substantial part of agricultural waste. The global annual production can achieve 2802 million tons [35, 36].

Food crops are currently used to produce bioethanol and biodiesel, threatening food security and land use. Seeking viable solutions to produce biofuels to substitute fossil-based fuels without affecting the food supply, research results showed that organic wastes can be used to produce biofuels and chemical products of use in the industry. Although the results are promising, more research and development are needed to enable implementation on a real commercial scale. Engagement of the fuel industry, financial funding, and global incentives are necessary to drive these projects ahead.

In Brazil, irrespective of the achieved advancements in the production of bioethanol and biodiesel, the production of biodiesel is not enough for the internal consumption. More efforts are needed to seek new feedstock to increase biodiesel production, improve and update the technology for converting biomass and agriculture and food wastes to biofuels, develop and implement advanced generations of ethanol and biodiesel, and valorize sewage and organic urban residues to enhance energy security and efficiency.

4.1 Production of Bioethanol

Brazil has a long and evolving history with bioethanol, transitioning from a program focused on reducing petroleum dependence to a producer of sugarcane-based ethanol. Key milestones include the introduction of mandatory blending with gasoline in the 1970s, the Pro-álcool program, and the development of flex-fuel vehicles. Brazil's ethanol and biodiesel production reached a record in 2023, totaling about 43 billion liters [37].

The sugarcane bagasse resulting from manufacturing first generation ethanol together with other organic matters is used for generating electricity or producing second generation ethanol. Also, Brazil has a significant increase in corn-based ethanol production, growing from 40 million liters in the 2013/14 to an estimated 8.2 billion liters in the 2024/25 season. Corn ethanol is expected to account for a growing share of Brazil's total ethanol production, potentially reaching one-third by the 2025/26 season. Available data shows that bioethanol production from sugarcane is superior to that from corn, showing ethanol yield in L/ha of 9000 against 4400, while the crop yield in ton/ha is about 80–100 against 12 from corn. Corn produces significantly more ethanol per ton than sugarcane, yielding approximately 400–455 liters per ton of grain, whereas sugarcane produces around 70–85 liters per ton. While corn provides higher efficiency per ton, sugarcane requires less intensive processing. Corn allows for year-round production in Brazil using second-season crops. Corn ethanol production is increasing rapidly, with capacity expected to rise to 14 billion liters by 2033 in Brazil [38, 39].

The ethanol can be produced from different organic matters and agricultural wastes, including sugary, starchy, lignocellulosic and algae feedstocks, using different processing routes such as direct fermentation in the case of sugary feedstock, while in starchy feedstock, enzymatic hydrolysis is used and the product is then fermented to produce ethanol. In the case of lignocellulosic feedstock, the material is subjected to pretreatment, enzymatic hydrolysis to produce sugary matter, which is then subjected to fermentation to produce ethanol. Alternatively, the lignocellulosic feedstock is subjected to thermal treatment (gasification) to produce syngas, which is catalytically converted to produce ethanol. Algae feedstock is conditioned and pretreated before being subjected to fermentation to produce ethanol. These processes are shown in Figure 3.

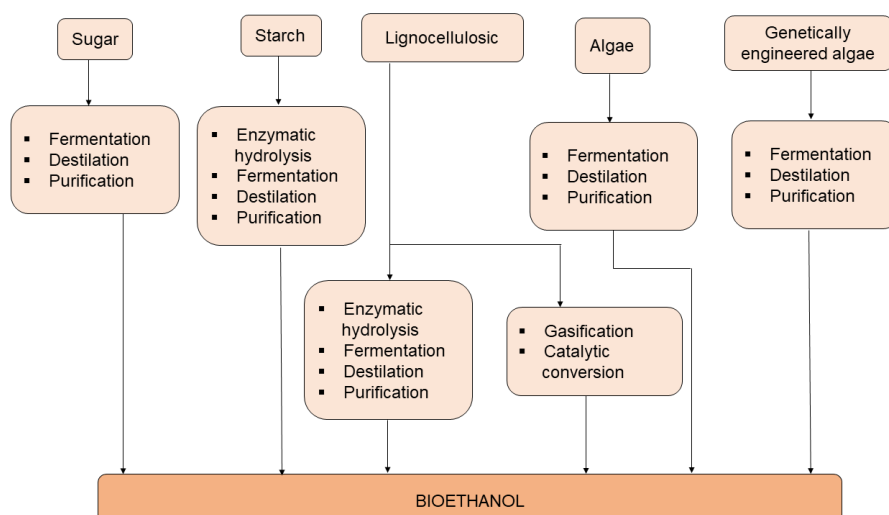


Figure 3. Possible routes of bioethanol production [40]

Sarkar et al. [41] presented a review on bioethanol production from agricultural wastes and showed that these wastes are an attractive option because they are cheap, renewable, and abundant, but several challenges and limitations need to be addressed besides reducing costs.

Zabed et al. [42] reviewed renewable sources to produce bioethanol, analyzed and commented on the perspectives and technological progress, focusing on the biomass sources and the associated technological approaches. Harshwardhan and Upadhyay [43] presented a study related to the conversion of biomass and agricultural waste and upgrading their economic value. Swain et al. [44] reviewed biofuel production from rice and wheat straw and showed that they are attractive materials to produce bioethanol. Rice straw global annual production is about 731 million tons, while in Asia the yearly production is about 668 million tons of straw, which can produce about 282 billion liters of ethanol. Global wheat straw produced annually is about 850 million tons, which can produce about 120 billion liters of bioethanol annually, but several challenges and limitations need to be addressed. Mohanty and Swain [45] investigated the use of corn and wheat for bioethanol production, evaluated the technology, its challenges and discussed the implications and impacts of the food and fuel interaction.

Production of ethanol from lignocellulosic materials is still in the development stage. Lignocellulosic agriculture residues were shown to be viable to produce bioethanol by using available techniques since they are abundant and of low cost [46]. Kannah et al. [47] reviewed food waste valorization and highlighted their physical and chemical characteristics and methods of pretreatment to produce biofuels and other products. Khan et al. [48] investigated biofuels, generations, advantages, and challenges besides the social and environmental impacts. Ambaye et al. [49] reviewed the technologies and challenges for using agricultural wastes and other biomass residues for producing biofuels and commented that the use of agro-wastes and microalgae to produce biofuels is an attractive and challenging process.

Mishra et al. [50] studied the ideal conditions to produce bioethanol from fruit wastes. Lee et al. [51] used thermo-chemical conversion on agricultural waste to obtain energy and other products. Awasthi et al. [52] analyzed biorefinery opportunities as a viable solution for waste minimization and discussed the integration of different treatment approaches as well as the resulting economic and environmental impacts. Babu et al. [53] reviewed the agricultural wastes for producing the second-generation biofuels and natural fertilizers. Saravanan et al. [54] reviewed recent advances in biofuels production from lignocellulosic biomass and highlighted recent advances in pretreatment methods. Rahimi et al. [55] reviewed the thermo-chemical conversion processes for turning agricultural wastes into biofuels. The characterization techniques for feedstock and the generated biofuels were also addressed. Bender et al. [56] reviewed the processes of bioethanol production from food waste and commented that the pretreatment is the biggest barrier to bioethanol production from food waste. Sarangi et al. [57] and Sonu et al. [58] reviewed agro-wastes and biomass, its conversion to sustainable energy, and discussed the biorefinery solutions to generate bioenergy. The authors also highlighted the associated challenges to be addressed.

Cherwoo et al. [59] presented a review on biofuel generations and possible implications and highlighted the challenges facing their commercialization. Shukla et al. [60] reviewed the production of bioethanol from agricultural wastes and discussed the possible technologies to produce bioethanol from these wastes. Samantaraya et al. [61] and Jamil et al. [62] reviewed bioethanol production from agricultural wastes, possible uses of bioethanol and its role in the green economy. The authors critically analyzed the various agricultural wastes, pretreatment and fermentation techniques used. Bioethanol produced from food residues was investigated by Abreu et al. [63], who commented

that the processes used were efficient, and viable. Ansari et al. [64] provided a review on food wastes, methods of pretreatment, sustainability, scalability and the impact of the technological integration on the valorization of food residues. Khaswal et al. [65] provided an overview of various agricultural and food wastes utilized in solid-state fermentation. The review encompasses a diverse range of microbial commodities and the possible yielding of a variety of industrially important enzymes. Aguiar et al. [66] analyzed how biofuel policies have affected the ethanol industry and how this industry has contributed to GHG emission reduction. The authors found that the use of ethanol fuel allowed significant emission reductions of 39–46% compared with gasoline.

Bioethanol production efficiency, productivity, and cost vary significantly based on feedstock type (sugary, starchy, lignocellulosic, or algae) and the processing route utilized. While first-generation feedstocks (sugary/starchy) currently dominate global production due to low processing costs and high maturity, second-generation (lignocellulosic) and third-generation (algae) feedstocks offer better sustainability profiles despite higher production costs. A comparative analysis is done on methods and feedstock to produce bioethanol with a focus on productivity, efficiency and cost.

In the case of sugary feedstock (1st generation), the process used is direct fermentation, where sugars (glucose, fructose) are directly converted by yeast. The efficiency is the highest among all types because no complex hydrolysis is needed. The productivity is with low processing costs due to the maturity of technology. The process of starchy feedstock (1st generation) involves starch gelatinization, enzymatic hydrolysis to convert starch to sugar, and then fermentation. The efficiency is high, about 90–95% theoretical yield of sugar to ethanol. The productivity/cost is high due to the mature technology, which makes it competitive (0.34–0.47 USD/L), but dependent on grain prices. The lignocellulosic feedstock (2nd generation-agro waste) requires expensive pretreatment (acid, steam, or enzymatic) to break down lignin-hemicellulose structures, followed by simultaneous saccharification and fermentation (SSF). The efficiency is lower than in the case of the first generation due to challenges in fermenting pentose sugars and inhibitor formation. The cost is high due to complex pretreatment and enzymes, though feedstocks are low-cost. Costs are roughly 0.21–1.25 USD/L. The process for handling algae feedstock (3rd generation) is similar to lignocellulosic but often with simpler pretreatment; however, macroalgae requires specialized microbes to ferment specific compounds like alginate. The efficiency is due to the high carbohydrate content (up to 60%), providing a strong yield potential. Currently, the high cost of water, nutrients, and harvesting makes it less economically viable (0.76–0.91 USD/L) [67, 68].

Remarks

In 2024, Brazil produced 36.83 billion liters of ethanol, or about 4.4% more than in 2023, of which about 7.7 billion liters were obtained from processing corn [69]. To cope with the actual and future demands of bioethanol without land use change, threat to food security, or deforesting to increase land destined to energy production, Brazil needs to invest heavily in research and new technologies to use agriculture and food wastes and organic residues from MSW to produce second-generation ethanol, biodiesel, and gaseous fuels. Also, it is essential to invest in developing the third- and fourth-generation biofuels to reduce the stress on land and water use and make more land available for food production. This can help reduce and stabilize the local food prices and create jobs and income besides the reduction of emissions.

4.2 Production of Biodiesel

The use of food crops to produce biodiesel can create serious insecurity in the food sector besides competing in land use and water supply. Extensive research and development were focused on in recent years to find substitute feedstocks such as used oil and fats, among other materials, to reduce the pressure on food crops. The examined alternatives were found insufficient to cope with the ever-increasing demands for biodiesel fuel. Agricultural and food wastes can help solve the problem of the production of biodiesel (2G) as they have several benefits, including low cost, renewability, and high generated quantities [70, 71]. Figure 4 shows a line diagram for the production of biodiesel where both vegetable oils and animal fats are used.

Biodiesel significantly contributes to Brazil's energy transition and promotes sustainable practices. Furthermore, biodiesel production supports rural development and provides economic opportunities for family farmers. In 2023, Brazil's biodiesel production reached over 7.5 billion liters, or about a 19% increase in relation to 2022. Soybean oil is the main feedstock used to produce biodiesel in Brazil and accounts for about 70% of the raw materials. One metric ton of soybeans typically produces approximately 180 to 220 liters of biodiesel. This process involves crushing soybeans to yield roughly 18–22% oil, which is then refined and converted through transesterification. Soybean oil remains a dominant feedstock for biodiesel. The biodiesel industry also supports the economic development via the Social Fuel Stamp, which encourages sourcing from family farming. Production of biodiesel from one ton of fats (such as waste cooking oil, tallow, or lard) generally yields approximately 800 to 1,000 liters of biodiesel via transesterification [72, 73]. Other materials like used cooking oil, animal fat, palm oil, and agricultural wastes also are used for biodiesel production to reduce reliance on soybean oil and address potential environmental issues associated with its production [73, 74].

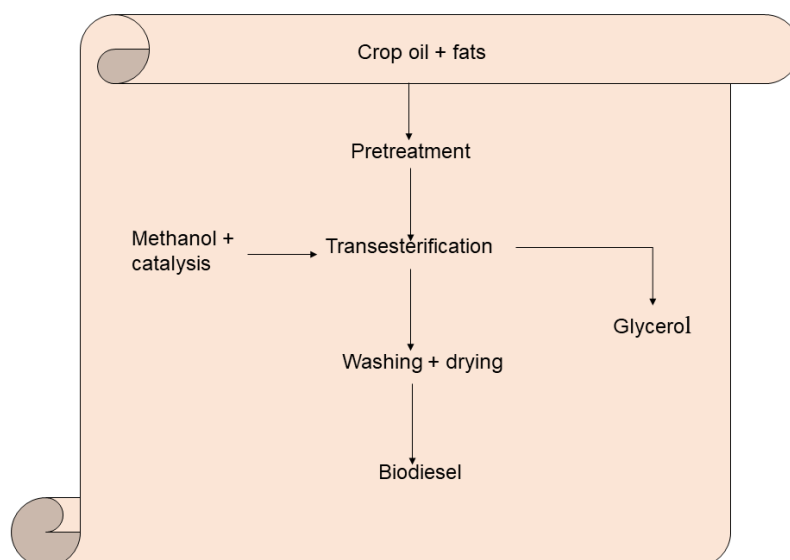


Figure 4. Biodiesel production from crude vegetable oil and animal fats [71]

Biodiesel production using edible oils seriously conflicts with food issues like supply and security, besides the high cost of both fuel and food and possible changes in land use. Used oils and animal fats are valuable and viable alternative options for biodiesel production without affecting the food supply [75]. The review by Oliveira et al. [76] showed the environmental and economic issues of biodiesel in Brazil. Sampaio et al. [77] analyzed the strategies conducted for the biodiesel sector and commented the importance of both research networks and innovations for the biodiesel industry.

Biodiesel production faces cost challenges, technical problems such as cold start performance of engines, and other post-production issues, besides possible interference with food supply. Manafa et al. [78] discussed these challenges and provided possible directions for research and developments. Aron et al. [79] provided a review to evaluate the production of biofuels, the net GHG emissions, and energy efficiency of the four generations of biofuels. The authors commented on the viability of the production processes and highlighted the implications and the associated challenges.

The third and fourth generations are under development and utilize algae and its genetically modified versions [80]. Azadbakht et al. [81] investigated the use of agro-wastes and animal fats to produce biodiesel and the results showed that the process is viable and profitable. Ong et al. [82] investigated the use of ionic liquids for biodiesel production from agro-residues and microalgae and discussed the process improvement and challenges of implementing this technology in biodiesel production. Mathew et al. [83] provided a study on recent advances in biodiesel production, advancement in technology and production processes, challenges, impacts and limitations, and possible solutions. Babadi et al. [84] presented a study focused on recent technological processes for biodiesel production, including implementation difficulties, expansion limitations, technical and production cost challenges, interference with food production and consumption, and possible opportunities. Silva et al. [85] analyzed the perspectives of biodiesel in Brazil and highlighted the environmental, social and governance aspects associated with biodiesel production and the accelerated growth of biofuels production due to the incentives and investment policies implemented in the last decades [86].

Glycerol is generated as a byproduct of the biodiesel industry at the rate of 10% of the produced biodiesel. The accelerated global biodiesel market makes crude glycerol a cheap substrate for ethanol production. Studies using crude glycerol to produce ethanol showed lower production costs in comparison with other substrates [87]. Research and development efforts were dedicated to investigating techniques and methodologies to produce bioethanol from crude glycerol, as in Jitrwung and Yargeau [88], who showed that crude glycerol can be used for the production of hydrogen and bioethanol, two biofuels that can help decarbonize the transport sector and integrate glycerol into bioethanol industries [89]. Chilakamarry et al. [90] investigated the production of bioethanol from glycerol waste and concluded that the process is technically and economically viable. Shan et al. [91] investigated direct hydrogenolysis of glycerol to ethanol, showing high ethanol selectivity, good reusability, and a high ethanol yield of 84.5%. Mehariya et al. [92] developed a fermentation process with crude glycerol, and irrespective of the impurities and shortage on nitrogen, the process showed good fermentation results. Since crude glycerol is cheap and abundant, the developed process shows that the crude glycerol can be an attractive alternative for biofuel production.

Remarks

Although the current biodiesel production is high, the product is not sufficient for the Brazilian internal demands. New investments in research and technologies are needed to expand production of biodiesel from agricultural and food wastes besides dominating the technology for the third and fourth biodiesel generations. With the increase of biodiesel production, there will be a corresponding increase of glycerol, and hence Brazil needs to invest in research and technologies to convert glycerol to bioethanol and other possible products. This will create new markets, jobs and income besides strengthening the circular economy and reducing environmental impacts.

Brazil produced 9.07 million m³ of biodiesel in 2024, about 20.4% more than in 2023, considering that the total world production of biodiesel in 2021 was about 151 million m³. The approval of the law “Fuel of the Future” stimulated the biodiesel market in Brazil, and the production can achieve 15 billion liters in 2030 [37, 93].

4.3 Production of Biogas and Biomethane

Anaerobic digestion is the process used to convert organic matter to biogas. A typical line diagram for the production of biogas, production of biomethane and biofertilizer is shown in Figure 5. The generated biogas is composed of methane, carbon dioxide and small amounts of other gases and impurities. Biomethane is obtained from biogas by removing the carbon dioxide and impurities. Both fuels are in use in the transport, residential and commercial sectors, besides contributing to a circular economy and mitigating GHG emissions [94].

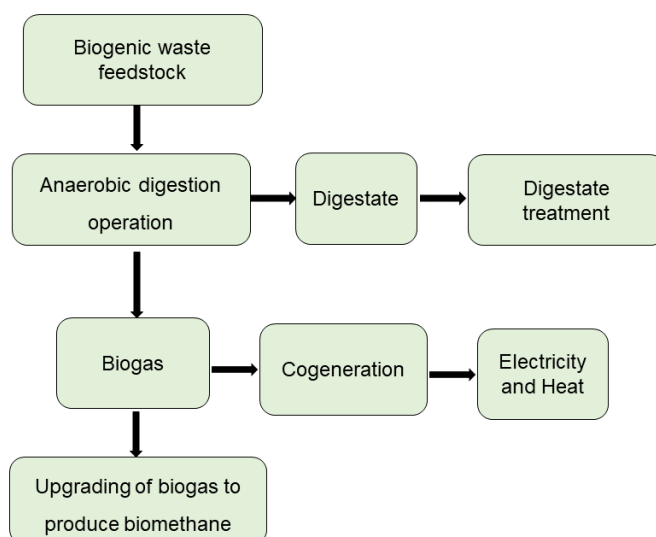


Figure 5. Production of biogas, biomethane and biofertilizer [94]

Biogas can be produced by the biodigestion of wastes from agriculture and food processing, wastewater, manure, and other organic matters, while the remaining inactive solids can be used as natural fertilizers. Biogas has a calorific value that varies from 16 to 28 MJ/m³, while that of biomethane can reach 35 MJ/m³. It is expected that biogas can have a significant contribution to the decentralized energy generation [95].

While Brazil has a big and growing biogas production sector driven by its abundant agricultural resources, only a small fraction of that potential is being used. Recent investments and policy changes are fostering growth in the sector. Biogas produced in Brazil in 2020, achieving about 1.8 billion Nm³/year, is mainly from waste treatment plants. Biogas from biomass can reach 82.58 billion Nm³/year. The Brazilian biogas potential can achieve annually about 84.6 billion Nm³, enough to substitute 70% of diesel consumption [96].

Chandra et al. [97] reviewed the biomethane production from corn, wheat, rice and sugarcane by using the biological route and concluded that generation of biomethane from lignocellulosic agricultural crop waste is cheap and environmentally correct. Cucchiella and D’Adamo [98] evaluated the economic performance of using biomethane for cogeneration or as vehicle fuel. Biomethane used as vehicle fuel presented good financial results and significant environmental gains.

Biogas upgrading improves the biogas users’ acceptance and increases its energy content. Current upgrading methods are expensive and have high energy consumption, which limits their application commercially [99]. Adnan

et al. [100] reviewed the available upgrading technologies and discussed the challenges associated with upgrading biogas. In their study Garcia et al. [101] evaluated agriculture and food wastes as feedstock for producing methane and commented that the production is sensitive to the fraction of organic matter in the residues. The investigation by Nascimento et al. [102] showed that landfill biogas is little explored in Brazil, only about 7 to 20% of the landfill gas produced is used. Prussi et al. [103] investigated the upgrading technologies for biogas and highlighted its concurrence with natural gas. Zhu et al. [104] addressed issues related to sustainability, policy instruments and farm intensification. Beniche et al. [105] examined the effects of co-digestion of a mixture of agricultural wastes on the production and thermal quality of the generated methane. The results showed that anaerobic co-digestion is effective and viable for treating the investigated mixtures. Sun et al. [106] estimated the biomethane production from Chinese crop wastes and the results showed a biomethane yield of $82.25 \times 10^9 \text{ m}^3/\text{year}$ in 2019. Dar et al. [107] reviewed the use of lignocellulosic agricultural wastes to produce biogas and discussed some relevant techno-economic aspects and policy issues. Deena et al. [108] reviewed the enhancement methods to produce biogas by co-digesting food residues and activated sludge. Favorable findings were obtained, including cleaner biogas, solutions for critical wastewater treatment, pollution problems, and increased energy demands, among others. Silva et al. [109] evaluated the viability of fueling vehicles with biomethane from landfills. The results indicated that this action can reduce emissions by 29.14 trillion tons of CO_2eq .

Enriched biogas can substitute natural gas and can be compressed and used as liquid biogas for exportation [110]. Devi et al. [111] reviewed waste treatment and management, use of wastes for recovering materials and obtaining energy, and economic, technical, and environmental impacts of production of biogas. Quevedo-Amador et al. [112] reviewed the advances in liquid biofuel production from waste biomass, their advantages and drawbacks, highlighted the conventional biochemical and thermochemical biomass conversion technologies to produce bioalcohols, and bio-oil. Alengebawy et al. [113] reviewed the production of biogas from agricultural wastes, utilization of biogas in urban and rural areas and the impacts on sustainability.

Remarks

In 2025, Brazil's biogas sector is experiencing rapid expansion, with 1,633 plants total and 248 new units added in 2024, producing 4.7 billion cubic meters annually, according to a May 2025 report from the World Biogas Association. Landfills and waste treatment plants are major sources, with significant investment shifting toward upgrading biogas to biomethane, which is projected to surpass electricity in production value by 2026. While landfill biogas is strong, the overall industry focuses on using urban organic waste, sewage, and agricultural residues, with a total potential of 84.6 billion Nm^3/year . The Future Fuels Law is driving the market, with mandatory biomethane certificates for producers and importers, significantly boosting investment in waste-to-biomethane projects [114, 115]. The challenges to the growth of the biomethane production sector include research and technological advancements, investments in infrastructure, and robust public policies to support the sector's growth besides investing in improving available technologies to handle co-digestion of food and agricultural wastes combined with other wastes.

4.4 Production of Biohydrogen

Biohydrogen can be generated from biomass by a variety of processes, including thermochemical, biological, and electrochemical as shown in Figure 6.

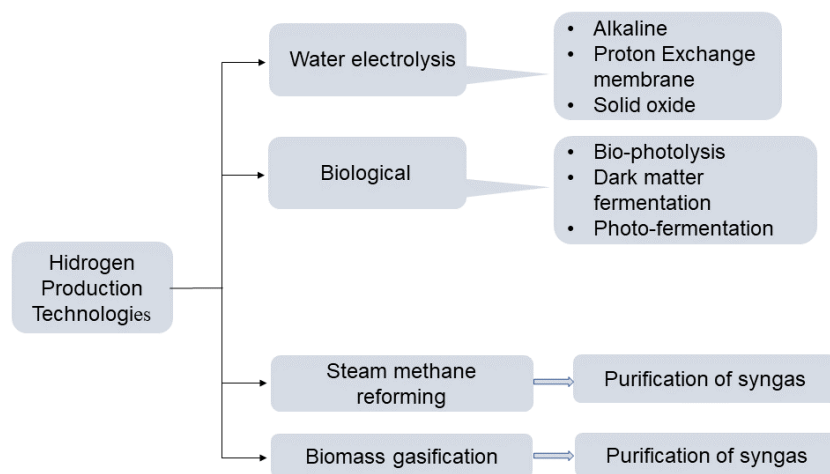


Figure 6. Organizational chart for hydrogen production routes [116]

The thermochemical processes include gasification and pyrolysis and steam reforming. In these thermal methods the biomass is converted into hydrogen-rich syngas, where hydrogen is subsequently obtained by purifying the syngas. In the case of pyrolysis, the biomass is heated without oxygen to produce gaseous fuels. Steam reforming is a viable technique for producing hydrogen, but its applicability is limited. The biological routes for producing hydrogen involve the use of biochemical processes such as fermentation and digestion by certain organisms to break down water molecules and produce biohydrogen. In the electrochemical routes, electricity is used to convert biomass into hydrogen. Since most biomass consists of water in varying percentages, it is acceptable to use water as an example of biomass. Electrolysis is a method for producing hydrogen from biomass at low temperatures, while in microbial electrolysis, the hydrogen production is via the catalytic action of microorganisms [102, 103]. The hybrid methods include integrated biomass gasification and hybrid pyrolysis/plasma reforming. These techniques are still under development for biohydrogen production. The biological techniques are attractive but need more research and development to enhance their productivity [117].

Currently, approximately 95% of global hydrogen production relies on steam methane reforming, which emits between 9 and 12 kg of CO₂/kg H₂ produced. Green hydrogen produced through water electrolysis powered by renewable energy sources results in near-zero GHG emissions, with total emissions of approximately 0.4–2.4 kg CO₂ eq/kg H₂. Biohydrogen derived from biomass gasification achieves net emissions of around 0.2 to 3.0 kg CO₂ eq/kg H₂. These findings underscore the environmental benefits of transitioning to green and biohydrogen pathways [116].

Brazil has a significant potential for biohydrogen production due to abundant agricultural residues. Studies indicate that various biomass sources, including sugarcane bagasse, vinasse, cassava wastewater, and urban waste, can be utilized to generate biohydrogen through fermentation and other biological processes. Biohydrogen production processes, primarily dark fermentation, photo-fermentation, two-stage systems, and biocatalyzed electrolysis, offer renewable alternatives to fossil-based hydrogen production, operating under mild conditions (ambient temperature and pressure) and utilizing organic waste as feedstock. While generally exhibiting lower production rates than conventional steam methane reforming, these biological routes provide significant environmental benefits, including GHG emission reductions of 57–73%. Biohydrogen production processes vary significantly in efficiency, with dark fermentation offering high production rates but lower yields, while photo-fermentation and biophotolysis provide higher theoretical yields but slower rates. Microbial electrolysis cells (MECs) boast the highest efficiency (>90%) by combining biological and electrical energy, though they require external power. Biohydrogen production processes vary significantly in productivity, with dark fermentation (DF) generally offering the highest hydrogen production rates and scalability, while photo-fermentation and two-stage processes achieve higher stoichiometric yields but lower rates. MECs show the highest efficiency (>90%) but are currently less mature [118].

Singh and Das [119] reviewed the hydrogen production from biomass by the dark fermentation process. Factors for improvement include metabolic engineering of microorganisms, temperature, pH, culture medium, composition, hydrogen partial pressure, and hydraulic retention time. Galvan et al. [120] revealed that solar PV, wind and gas compose the main pillars for energy transition in South America. Soares et al. [121] evaluated the potential of biomass waste for producing biohydrogen, ethanol and chemical elements in Brazil, with an estimated yearly production of biohydrogen of 96.14 million m³. Chantre et al. [122] studied different aspects of the Brazilian hydrogen economy development and showed that there is a need to create market demand, develop standards and regulations and enhance R&D support. Goria et al. [123] investigated the feedstock and methods of biohydrogen production, barriers and challenges and possible solutions. Brar et al. [124] reviewed the advances in genetic engineering to enhance hydrogen yield. Pal et al. [125] examined biohydrogen and advancements in technology, including biological processes, costs, challenges of the different hydrogen production methods and future research and development opportunities. Piovani et al. [126] commented on the need to create industrial incentives, public policies and more investments and research to enhance green hydrogen production in Brazil. Garlet et al. [127] conducted a study to identify green hydrogen production in Brazil, the challenges, opportunities, investments, and stimulation. De Figueiredo et al. [128] conducted a study on green hydrogen production routes and possible applications in the Brazilian market.

Goveas et al. [129] investigated the enhancing factors for hydrogen production and commented on the use of microbial immobilization and nanotechnologies for improvement of hydrogen yield. Karthikeyan and Velvizhi [130] presented a review that addresses the influence of non-material on the production of biohydrogen. Nanoparticle interactions with bacteria, substrate, and enzyme changes were investigated. Rath et al. [131] reviewed the various biomass and biohydrogen production techniques from biomass. Each form of production's methods, advantages, and disadvantages have been investigated. Nanoparticles can enhance the biohydrogen yield, but further research is required to demonstrate practical applications and commercial viability. Silva et al. [132] evaluated the viability of producing green hydrogen from sugarcane biomass and its possible integration within the existent energy infrastructure. Santos et al. [133] provided an overview and a technical analysis of using peach industry residues to produce biohydrogen. The generated biogas and energy were evaluated as 1 million m³/year and 8.63 MWh/year, respectively.

Remarks

Biohydrogen research and developments in Brazil are in the development stage. Recent governmental efforts to strengthen the hydrogen industry included approving the National Hydrogen Program (PNH2) [134], with a focus on the decarbonization of the energy sector, valorization of the national technology and the development of a competitive industry. Government policies like the “Green Hydrogen Law” and the PNH2 are driving investments and innovation in this area. Incentives for international collaboration and mutual development programs and exchange of experiences are welcome to speed up the development process on the national level. Brazil needs to invest more in research and development and create public policies to encourage biohydrogen production. These actions, along with others, can provide the foundations necessary to develop the national technology and create jobs.

5 Contribution of the Pulp and Paper Industry Waste-to-Energy Transition

The residues of some industries are significant and cause serious damage to the ecosystem besides the tremendous losses of materials and energy. The pulp and paper industry is chosen to be analyzed and indicates its potential to help achieve a sustainable energy transition in Brazil. The pulp-paper industry generates huge amounts of sludge and waste with a strong negative impact on the environment. Landfilling or incineration of these wastes is restricted, and other management processes are encouraged, especially those leading to energy recovery and/or reuse of material.

Pulp and paper industries contribute about 26% of MSW in landfills, besides being responsible for 4% of the global energy use. Brazil’s pulp sector experienced strong recovery and significant capacity expansion in 2024 with a focus on packaging and hygiene products and achieved around 25.2 million tons driven by e-commerce packaging and hygiene products. Key growth drivers for Brazil included increasing exports, achieving 18.6 million tons, and significant investments in eucalyptus plantations [135, 136].

Paper industry waste, primarily biomass sludge and used paper, offers significant potential for the energy transition by generating bioenergy (biogas, biohydrogen and heat) through valorization, reducing reliance on fossil fuels and cutting GHG emissions, especially when paired with increased recycling [137].

The current situation offers vast opportunities for improvement, considering that waste biomass (sludge, black liquor) can be converted into biofuels (bioethanol, biogas) or used directly for heat/power, boosting energy self-sufficiency. By utilizing waste for energy reduces the need for external fossil fuels, lowers emissions of the sector and saves water. Recovering fibers from used paper, reduces landfill waste and the demand for virgin wood, supporting sustainability and a circular economy. Introducing advanced waste valorization technologies like anaerobic digestion and biorefineries can turn the problematic sludge into valuable energy products. Higher recycling rates significantly lower the overall energy footprint, while implementing cleaner technologies such as better bleaching, water treatment, and energy efficiency mitigates direct pollution [136, 137].

Biorefinery beneficiation of pulp-paper industry sludge can produce a variety of products such as biohydrogen, bioethanol, and biodiesel. Adequate thermochemical processes can produce bio-oil, hydrochar, biochar and activated carbon, while recycling makes it possible to incorporate it into cardboard and construction materials [137]. Fahim et al. [138] analyzed the contents of the paper and pulp sludge and found that it is rich in phosphorus and magnesium. The application of sludge to the soil improves soil fertility, biomass production, and plant growth. Mendoza Martinez et al. [139] discussed the techniques used for handling pulp mill sludge, including hydrothermal carbonization to convert wet organic substrate into a value-added source of materials such as hydrochar, which can be used in energy, soil and adsorption applications.

Recovering energy from wastepaper sludge can contribute to energy independence and improve waste management, increase renewable energy generation, and reduce emissions [140]. Lipiäinen et al. [141] investigated the response of the paper sector to the decarbonization in Finland and Sweden. It was found that significant improvements were achieved, including energy efficiency, and partial reduction of fossil fuel use and CO₂ emissions besides the increase of renewable electricity generation. Vilarinho et al. [142] summarized the investigations focusing on binders and mortars production from wastes for building applications, novel materials for high-added value applications and highlighted the barriers and future prospects. Mendoza-Martinez et al. [143] investigated fast pyrolysis to produce bio-oil for the pulp industry. Pyrolysis produced biochar; gases, which were used to heat the fluidization air; and bio-oil (heavy and light), where the heavy bio-oil showed a yield of 30%, adequate for fuel applications. Kumar et al. [144] delivered a review of pulp-paper industry sludge management approaches, reuse in building materials, developing super capacitors, cardboard, biofuels, and cellulose, among others. Kirthika et al. [145] mentioned that paper production generates paper mill sludge as a by-product of about 27.5 million tons by 2050. The authors evaluated the current sludge management practices and valorization opportunities such as energy recovery, material reuse, and biofuel production. Gonçalves et al. [146] evaluated the properties of sludge from pulp mills and the use of solar energy to dry the sludge from an initial concentration of 21% to 95.5%, suitable for sludge incineration.

Remarks

Waste from the pulp and paper industry, such as sludge, lignin, and fibers, transforms these offensive wastes into high-value products like biofuels (biomethane, ethanol), biochar, bioplastics, and construction materials (cement, bricks). Using techniques like hydrothermal treatment, anaerobic digestion, and biorefining, these waste streams are converted into renewable energy and chemical feedstocks, supporting a circular economy, decreasing landfill reliance, and reducing methane emissions [144].

Valorizing these wastes in Brazil is a critical component of the country's growing circular bioeconomy, focusing on transforming solid residues and sludge into bioenergy, agricultural inputs, and construction materials. This industry increasingly utilizes biomass waste and black liquor for co-generation, boasting renewable energy rates above 85%. Key challenges involve managing potential pollutants in effluent sludge (such as organochlorines from bleaching) and the high costs associated with drying and transporting residues for agricultural use [135, 136].

6 Contribution of the Construction and Demolition Industries' Wastes

Globally, the building industry extracts about 30% of the natural resources, generates about 25% of waste, and accounts for 40% of global carbon emissions, which ranks it as a significant contributor to global environmental challenges. Traditionally the building industry adopted a linear economic model, which resulted in the current situation of high energy and materials consumption accompanied by huge amounts of waste. A shift towards the circular economy model can improve the material and energy use in the building industry [147].

In Brazil, the CDW is significant, accounting for about 38.2% of the total MSW, with only about 7.2% by weight recycled and returned back to construction activities. The primary focus of CDW in Brazil is centered on material recycling to conserve natural resources and reduce environmental impacts. The main role of CDW management is shifting from a linear to a circular economy model, which indirectly supports energy transition by reducing the energy needed for new material production. Brazil generates approximately 45–70 million tons of CDW annually, most of which is highly recyclable. Recycling CDW into aggregates for use in public works (e.g., road paving) is a key strategy. This preserves natural resources like sand and gravel. Proper management and recycling offer economic benefits, such as reduced disposal costs for municipalities and the creation of jobs and income in the recycling industry [148, 149].

According to study [149], Brazil produced approximately 48 million tons of CDW in 2021. In 2024, Brazil's construction sector continued to generate significant amounts of CDW, representing a major environmental challenge, with estimates of the order of 48 million tons annually, though only a small fraction (around 7–15%) was recycled, highlighting vast potential for resource recovery and circular economy practices, besides the growing interest in sustainable solutions such as waste-to-energy and recycled aggregate use.

Ginga et al. [150] and Chang et al. [151] reviewed material recovery, production, recycling of CDW, limitations, research gaps and the potential of green building information modeling (BIM) research were discussed. The authors recommended more research focusing on integrated design analysis. Goel et al. [152] conducted a study focused on using paper mill sludge compost for the production of more porous and lighter bricks for light building construction. Experimental results showed that these bricks were safe to use in regular applications as infill walls. Rahla et al. [153] provided a review on criteria for building elements selection according to the circular economy. The authors commented that little has been concretely achieved regarding a shift to the circular economy, outlined the current adopted criteria and highlighted the need for an approach to attain higher circularity levels. Hentges et al. [154] presented the current context of public policies in the Brazilian construction industry and proposed initiatives to introduce principles of the circular economy. The authors recommended more initiatives to effectively implement a circular economy in the Brazilian building industry. Abulebdah et al. [155] conducted a study to discuss CDW management and developed an integrative approach to help in the process optimization and demonstrated that integrative management strategies can enhance the process and reduce costs by about 52%.

Management of wastes from construction and demolition is challenging but also offers valuable opportunities, especially in rapidly urbanizing countries such as Brazil. The review analyses the CDW management practices in Brazil with reference to the United Nations Sustainable Development Goals (SDGs). Challenges specific to Brazil include logistical difficulties in remote areas, the economic feasibility of recycling practices, and environmental risks, especially in ecologically sensitive regions like the Amazon. The study identifies critical research gaps and proposes future directions for developments in emerging economies [156]. The building industry globally consumes huge amounts of natural resources, generates 20% of carbon emissions, and the management of its generated waste represents significant challenge. Recycling these wastes is economically viable and can avoid 37 kg of CO₂ equivalents per ton of waste besides helping to preserve natural resources and the environment (Javed et al. [157]). Vásquez-Cabrera et al. [158] analyzed the circularity in a multi-family building and concluded that recycled aggregate concrete can enhance the material circularity performance by about 42.82%.

Remarks

Valorization of CDW converts rubble, concrete, bricks, and wood into high-value construction materials, promoting a circular economy. CDW accounts for roughly 50% of urban solid waste in Brazil, with over 80% having recycling potential (e.g., concrete, brick, wood). However, only a small portion is recycled or reused (approx. 7.2%–15%), often restricted to non-structural uses like paving. Common valorization includes producing aggregate for low-cost bricks and sub-bases, aiming for a circular economy. These practices reduce landfilling and provide sustainable alternatives to natural materials [159].

7 Contribution from the Scrap Tires

The global generation of scrap tires is a major environmental challenge, with estimates ranging from over 1 billion to 1.5 billion tires generated annually, of which a significant portion, reported as over 40%, is still discarded in landfills, stockpiles, or via illegal dumping. Discarded tires do not easily biodegrade and present several severe environmental and health risks. Tire piles are highly flammable, and once ignited, they release toxic pollutants, including dioxins, polycyclic aromatic hydrocarbons, and fine particulate matter. Also, tires contain vulcanized rubber and chemical compounds like zinc and lead that can leach into soil and groundwater, contaminating ecosystems [144].

Laftah and Abdul Rahman [160] presented a comprehensive review of tire recycling technologies and applications to produce scrap tire oil which is attractive both economically and energetically.

Machin et al. [161] reviewed processes such as gasification and pyrolysis to convert scrap tire rubber to energy. Scrap tires generated in Brazil are about 20 million units, mostly discarded in landfills, but lately, increasing amounts are used as raw material for paving roads and construction of buildings. The implementation of these treatment processes in Brazil can help solve the disposal of waste tires, create jobs, and increase distributed generation of electricity. Czajczyńska et al. [162] in their review commented that pyrolysis of the waste tires can produce gaseous, liquid, and solid useful by-products. The gases have a high heating value of about 30–40 MJ/Nm³, but have a high concentration of SO₂, which needs cleaning methods to remove it from fuel gas. Asaro et al. [163] presented a literature review on rubber recycling by devulcanization, such as chemical, ultrasound, microwave, thermo-mechanical, etc., and concluded that thermo-mechanical devulcanization based on extrusion is more suitable for large-scale production.

Application of pyrolysis to scrap tires can provide a solution to this waste management and produce oil (that can be used as fuel) and char, which can be converted to porous carbon [164–166]. Gamboa et al. [167] investigated the tire pyrolysis process to produce oil, which can be used as a fuel. They estimated an annual production capacity in Brazil of 230 to 280 thousand m³. Mavukwana et al. [168] analyzed an integrated gasification combined cycle system fed from processing 518 ton/day of waste tires. The results showed an efficiency of 45.65% and power production of 89 MW. Martínez [169] reviewed scrap tire generation rates, regulating policies, and recovery processes used with a focus on pyrolysis and circular economy. Some challenges and opportunities were addressed. Dong et al. [170] used the life cycle assessment technique to determine the environmental impacts of tires along the stages from production to use, on-going trends, challenges, and future opportunities. Gao et al. [171], Doja et al. [172], and Wang et al. [173] provided reviews on tire char production, recycling routes and application, the productivity and characteristics of tire char, and explored the challenges and future opportunities. Zerin et al. [174] reviewed the pyrolysis of tires and the chemical activation of tires, and showed that tire pyrolysis can produce 30–65% oil, 25–45% char, and 5–20% gas, which makes them excellent candidates as fuel substitutes and energy storage.

Han et al. [175] provided a review that outlines the thermochemical techniques for recycling tires, including gasification, pyrolysis, and incineration, besides delving into the primary byproducts, including oil, gas, and char. The advantages and drawbacks of each process were discussed and the environmental impacts were commented on. Future research directions were also addressed. Wu et al. [176] conducted research to enhance the quality of fuel by incorporating hydrogen into scrap tire pyrolysis oil. The results showed improvement of the combustion process and reduction of carbon emissions. Ore and Adebisi [177] investigated the pyrolysis of waste tires and showed that the gases obtained can serve as a source of hydrocarbon gases. Şen et al. [178] investigated the upgrading of waste tire pyrolysis oil and the results showed acceptable performance and possible use as a fuel for diesel engines. Pei et al. [179] reviewed the status of research and development related to mechanical processes for recycling discarded tires, examining input variables, performance indicators and their relationships. By integrating advanced engineering knowledge with a good combination of the latest technology, it is possible to extract the reusable materials of discarded tires and convert them into reusable forms like energy and hydrogen. Borkar et al. [180] investigated the use of hydrogen and tire pyrolysis oil as fuels, discussed the research gaps, and recommended some future investigations. Yaghi et al. [181] provided a review that explores the latest advancements in the pyrolysis of discarded tires such as fixed and fluidized beds, along with key process conditions that influence the quality of the tire pyrolysis oil. The review also examined the co-pyrolysis of discarded tires with other waste plastics, such as polyethylene, polypropylene, and polystyrene to enhance the quality of the tire pyrolysis oil.

Remarks

Globally, over 1.5 billion waste tires are generated annually worldwide. Only about 42% are utilized in material recovery, and 15% for energy recovery, while the rest ends up in landfills.

Brazil has made significant efforts in valorizing discarded tires, with over 90% of unusable tires being recycled or repurposed, placing the country as a global leader in environmentally correct tire disposal. This success is largely due to a 2009 national policy requiring reverse logistics, where manufacturers are responsible for collecting and processing used tires.

The valorization methods in Brazil include using them as an alternative fuel in cement furnaces, crumb rubber generated from shredded tires is added to asphalt mixes, which increases road longevity, reduces noise, and increases safety. Also, discarded tires are increasingly used in construction, such as for creating flood-resistant walls (a durable, flexible alternative to concrete) and for soil erosion control. Advanced pyrolysis processes are being developed to convert tires into valuable components like fuel oil, gas, and carbon black. Challenges include a need for better logistical coverage. Additionally, some high-value recycling techniques, like pyrolysis, are not yet fully implemented on a wide scale [164].

8 Contribution of Municipal Solid Waste to Energy Transition

Adequate management of MSW is a challenging problem due to its contamination impacts and risks to public health. The dominant way of waste treatment, especially in developing countries, is landfilling and dumping in open sites. Irrespective of the fact that these treatments are inadequate, huge funds are spent on the transport and depositing of these wastes. Global MSW generation is projected to rise to 2.59 billion tons by 2030. Current global waste management practices include landfilling, which accounts for 37–40% of MSW disposed in various types of landfills, dumping and uncontrolled disposal account for 30–33% of waste, recovery and treatment (recycled or composted) accounts for about 19%, while 11% is incinerated [182].

MSW in Brazil can significantly contribute to the energy transition if managed and reused properly. Incineration and biogas production from MSW are key methods, where incineration can produce electric energy while biogas can be converted to biomethane for thermal and transport uses. Brazil does not have waste-to-energy plants in commercial operation fed by MSW, but has two pilot plant projects. The unit in Barueri, São Paulo is expected to consume daily about 825 tons of MSW and generate 17 MW, while that in Mauá, São Paulo can process 3000 tons/day of waste and generate about 77 MW. Ferreira and Balestieri [183] presented an analysis on electricity generation from biogas. Two combined cycle concepts were evaluated: gas turbine and incinerator, and gasification system. Comparative analysis demonstrated that the cycle with gasification was technically more appealing. Dalmo et al. [184] reviewed the energy recovery from MSW of São Paulo State, Brazil. Two technological routes were investigated: thermochemical and biochemical. The combination of incineration and anaerobic digestion showed a higher generated electricity of about 8,051,623 MWh per year.

The organic solid waste and food waste, which are usually used for animal feeding, incinerated, or disposed of in landfills, could alternatively be separated and adequately treated to produce biofuels, energy and chemical products. The processes involved need to be further investigated and improved to handle a mixture of different food waste [185]. Elkhaila et al. [186] reviewed the application of pyrolysis to convert food waste to biochar and concluded that the process needs more research and developments before it can be applied, especially for a mixture of different food wastes. Chen et al. [187] conducted a simulation study to compare waste production with reference to composition, methods of treatment, and impacts. The authors commented on the significant increase of the generated waste, decrease of dumping due to the restrictive measures adopted by most of the nations, and increase of recycling and energy recovery by different technical methods. Ayilara et al. [188] investigated composting as a route for the treatment and management of food wastes and analyzed the different composting methods, duration of composting, and present trends in composting and prospects. Djandja et al. [189] investigated the pyrolysis to produce biofuels from sewage sludge and evaluated the influencing parameters and the working conditions.

Usmani et al. [190] investigated MSW generation, characteristics, and methods of treatment to generate energy and recover materials and commented that about 2 gigatons of MSW are produced annually, of which nearly one third is not collected properly. The integration of MSW management systems can reduce both the volume and administrative costs and enhance the circular economy [191]. The reduction of generated MSW, incentives to recycling, and the conversion of MSW into energy, biofuels can help minimize pollution, and global warming [192].

Varjani et al. [193] examined the general characteristics of biomass and solid wastes, possible bio-routes and thermo-chemical methods for reutilization of MSW and highlighted innovations and existent challenges. Hoang et al. [194] and Khan et al. [195] reviewed the inter connection of MSW and circular economy, the benefits and limitations of the existing MSW management methods, the challenges, the opportunities, and the governmental policies to foster their implementation. Maluf Filho et al. [196] analyzed the obstacles and limitations to harness the biomass energy from MSW for distributed generation in Brazil. Over the years millions of tons were deposited

in landfills resulting in losing energy and reduction of GHG emissions. Rodrigues et al. [197] investigated using gasification and incineration for treating MSW and generating energy for three different municipalities. The authors commented that incineration was more viable for Campinas and Campo Grande. Dadario et al. [198] reviewed the advances in MSW treatment technology, energy generation from solid wastes and recommended the elaboration of regulations, public policies and incentives to increase generation capacity and improve MSW management.

An alternative way to recover energy from MSW is by using its carbon content to produce other chemical products. By this route it is feasible to reduce energy and emissions by replacing fossil-based compounds with those produced through this treatment route. The authors mentioned that there is only one demonstration plant fed by MSW in operation [199]. Senpong and Wiwattanadate [200] conducted a review study on waste-to-energy projects with the objective of determining their potential, barriers and challenges. Among the serious barriers to the implementation of this type of project is the strong rejection by society due to pollution and high initial costs. Yattoo et al. [201] presented an overview on global waste generation, composition, leachate formation, and their offensive impacts; management approaches to reuse material and energy content of MSW; and implementation of adequate methods for waste and leachate treatment.

Remarks

In 2024, MSW management in Brazil is heavily focused on implementing the National Solid Waste Plan (Planares), which aims to eradicate illegal dumping and accelerate the adoption of waste-to-energy and biogas technologies. Brazil generates approximately 81 million tons of MSW annually, and while 93% is collected, less than 3% of its energy potential is currently utilized, providing significant room for growth. Planares focus on closing, renovating, or replacing landfills with sanitary landfills to meet stricter environmental standards. Bioenergy recovery (including landfill gas-to-energy, biogas, and RDF) is being recognized alongside traditional recycling, accounting for around 11.7% of total waste treatment. Also, a new national strategy for organic waste was developed, focusing on composting to manage the high percentage of organic matter in Brazilian MSW. Despite these advancements, many municipalities still face challenges in replacing old dumps with modern, expensive waste-to-energy technologies, relying heavily on traditional sanitary landfills in the short term [14, 202–205].

9 Conclusions and Future Research Gaps

9.1 Conclusions

1. Energy transition is strongly connected to efficient and effective management of all wastes, including MSW, industrial and agricultural wastes, and eliminating dumping practices, besides intensifying the circular economy and recycling to reduce consumption of raw materials and energy. These actions will reduce impacts on the ambient, generate renewable energy, and reduce emissions and pollution of underground water and soil, besides creating jobs and income (SDG 3, SDG 8).

2. The adoption of residential and community photovoltaic systems is experiencing rapid growth, driven by falling technology costs, increasing electricity prices, and a global push toward decarbonization. Brazil has surpassed 3 million residential photovoltaic systems, with a total installed capacity of 42 GW in solar distributed generation by 2025. The segment grew from 1 million to over 3 million systems in just over three years. Decentralization and democratization of renewable energy generation and use attain (SDG 7).

3. Brazil holds one of the world's largest technical potentials for biogas, capable of meeting over 35% of the national electricity demand or 70% of the diesel fuel demand, utilizing agricultural and municipal waste to produce 2.3–4.7 billion/year as of 2021–2024. Currently Brazil utilizes less than 3% of its 84.6 billion annual potential, but the sector is growing rapidly, with a 15–16% increase in plants yearly. Challenges for wide expansion, include developing infrastructure for biogas transport and upgrading to biomethane, establishing a circular economy and reducing reliance on fossil fuels.

4. Biodiesel production in Brazil is based on soybean oil, which accounts for roughly 70–75% of total feedstock, followed by animal fat, while other vegetable oils like cotton, palm oil, castor oil, and sunflower have a big potential for possible use even on a small scale. Brazil produced over 9 billion liters in 2024, driven by a mandatory blending policy. The National Program for the Production and Use of Biodiesel (PNPB) mandates that a specific percentage of biodiesel be added to conventional diesel, reaching 20% by 2030. The sector's growth aims to reduce reliance on imported fossil diesel which is about 300,000 barrels of diesel daily, or 26% of the national total diesel consumption. Biodiesel in Brazil is capable of reducing GHG emissions by over 80% compared to fossil diesel.

5. Biofuels produced from agriculture and food residues can alleviate the pressure on food crops and avoid food and energy insecurities. Brazil uses agricultural and urban waste to produce biogas, biomethane, and second-generation ethanol. Key waste sources include sugarcane bagasse, animal waste, and agricultural residues, aiming to reduce emissions by up to 95%. While current biomethane production is small (around 365,000/day), the potential is estimated at 120 million/day. The government is pushing for increased investment to unlock this, with notable infrastructure projects for biogas-powered transport. Currently, only laboratorial studies are available on biohydrogen

and jet biofuel production from agriculture and food wastes, needing more research and development besides financial support. Another important feedstock for producing second-generation ethanol is the crude glycerol from the biodiesel industry to reduce environmental impacts and valorize the subproduct. There is a need for more investment and more research and development.

6. The pulp and paper sector in Brazil generates significant waste—including wastewater, sludge, ash, and liquor—that causes substantial negative impacts on the environment when improperly managed. These impacts include water contamination, soil degradation, air pollution, and social conflicts over natural resources. The demand for wood has led to the conversion of native biomes into Eucalyptus monocultures. This causes habitat loss, loss of biodiversity, and increased vulnerability to fire.

The Brazilian paper and pulp industry leverages its waste products to generate significant renewable energy, with the sector self-generating over 51% of its electricity needs. The primary energy benefit comes from burning black liquor and biomass residues. Key waste streams include black liquor, sludges, and ash, with ongoing research exploring their use for soil enrichment, asphalt, and energy recovery. Advanced mills are moving toward biorefining, extracting lignin from black liquor for use as a biofuel or raw material, which increases the overall energy efficiency and sustainability of the production process.

7. The Brazilian tire industry has significant environmental, economic, and social impacts, driven by a high-volume production market and a robust reverse logistics system designed to handle waste tires. While the industry has made progress in recycling, with over 90% of waste tires being properly disposed of, the sector faces challenges regarding environmental, social, and human health, particularly regarding improper disposal and the rise in low-cost imports. Waste tires in Brazil, with roughly 450,000 tons discarded annually, cause severe environmental and public health hazards due to improper disposal. As nonbiodegradable materials, they pollute soil and waterways. They also leach toxic heavy metals and organic compounds into ecosystems, including coastal areas. Energetic valorization of waste tires in Brazil primarily involves using them as alternative fuel in the cement industry (co-processing), driven by high disposal volumes (over 473,000 tons/year) and strict environmental regulations. Advanced alternatives like pyrolysis and gasification are emerging to produce fuel oil, gas, and energy, aiming for higher efficiency than incineration.

8. The energetic valorization of CDW in Brazil is an emerging field, driven by the need to manage over 70 million tons of annual waste, of which wood is often underutilized. The sector focuses heavily on recycling mineral waste into aggregates, 16–21 million tons/year. Recent studies indicate that blending wood waste from construction with biomass can yield efficient bioenergy. Wood represents roughly 15% of Brazilian construction waste, with an average composition of wood in construction waste varying from 2.21% to 37%. With new initiatives there is significant potential for transforming waste into energy that could supply 4 million homes.

9. The present review, although it treats the energy transition in Brazil, show that the methods, results, treatment routes, public policies and incentives can be extended to other developing countries similar to Brazil. Provision of excess renewable energy generation and use is lacking in some developing countries and to have a sustainable energy transition, it is important to satisfy the social pillar. Many developing countries are still facing problems associated with sewage and wastewater treatment, which, by adopting adequate public policies, awareness programs, and alliance with the private sector, besides using adequate technologies, can produce financial and ambient benefits. For the majority of developing nations, dumps and unchecked waste generation and deposition are key issues that require long-term solutions in order to guarantee a genuine and long-lasting energy transition. Conversion of organic and agricultural wastes to biofuels and other chemical elements can boost the local economy and promote a circular economy and distribute energy concepts necessary to attain the SDGs of the UN. In summary, the results and methodologies treated in this review can be applied to other developing countries similar to Brazil.

9.2 Future Research Gaps

1. Research results have shown that agricultural and food wastes can be transformed into biofuels and other useful commodities. More research and developments are needed to improve existing technology and integrate new unavailable technology to produce biohydrogen and jet biofuel for the internal market, besides investment in biorefineries of MSW to produce liquid biofuels and other products.

2. Improving public services of sewage treatment, wastewater treatment and provision of treated water to ensure better living conditions for all (SDG 3).

3. Brazil needs to invest effectively in local research and developments besides participating in cooperation projects with other nations to implement cheap and effective technologies for converting biogas to biomethane and biohydrogen (SDG 9).

4. Glycerol from the biodiesel industry is suitable to produce ethanol, bio-hydrogen and other products. Brazil should intensify projects and developments that could expand the biodiesel, methanol and bio-hydrogen fuels and create from glycerol additional liquid and gaseous fuels.

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Conceptualization, F.A.M.L. and K.A.R.I.; writing—original draft preparation, F.A.M.L. and K.A.R.I.; writing—review and editing, F.A.M.L. and K.A.R.I. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

ABEEOLICA	Brazilian Association of Wind Power and New Technologies
ANP	Agencia Nacional de Petróleo, Gás Natural e Biocombustíveis
BIM	Building information modeling
CDW	Construction and demolition waste
GHG	Greenhouse gas
MEC	Municipal solid waste
PNH2	National Hydrogen Program
PNPB	The National Program for the Production and Use of Biodiesel
PV	Photovoltaic
SDGs	United Nations Sustainable Development Goals
SSF	Simultaneous saccharification and fermentation